



SCENARIO CONFIGURATION

Cloud Provider	azure
Deployment Model	cloud_managed
Data Sensitivity	confidential
Maturity Level	intermediate
User Count	2,500
Budget Range	\$500k-\$1M
Timeline	12 months
Compliance	SOC 2 Type II, ISO 27001, GDPR, MiFID II
Strategic Partners	NVIDIA, Microsoft

ARCHITECTURE LAYER RECOMMENDATIONS — OVERALL SCORE 84/100

Cloud Foundation

 85/100

Confidence: 88%

RECOMMENDED

azure Landing Zone with Enterprise-scale architecture

azure is your preferred hyperscaler, offering mature AI/ML services, enterprise support SLAs, and native identity federation.

- MEDIUM** Cost: Landing zone setup is a one-time cost; ongoing management overhead is low
- LOW** Complexity: Managed by the hyperscaler; IaC templates reduce operational burden
- HIGH** Vendor lock-in: Cloud-native identity, networking, and billing tie you to the provider
- LOW** Time to value: Foundation can be deployed in 2–4 weeks using enterprise blueprints

Compute / GPU

 85/100

Confidence: 88%

RECOMMENDED

NVIDIA DGX H100 SuperPOD (on-prem) + Azure NC-series VMs (A100 / H100) with AKS for burst

High-throughput GPU compute is critical for model inference and fine-tuning. NVIDIA DGX provides best-in-class bare-metal performance for sensitive workloads.

- HIGH** Cost: GPU compute is the largest cost driver — reserved instances reduce by 30–40%
- MEDIUM** Complexity: Requires CUDA/driver management; managed node pools reduce this significantly
- MEDIUM** Vendor lock-in: GPU instances are portable across clouds; NVIDIA tooling is cloud-agnostic
- MEDIUM** Time to value: Quota provisioning for high-end GPUs can take 1–4 weeks on first request

Storage

 84/100

Confidence: 88%

RECOMMENDED

Azure Blob Storage + Azure NetApp Files

Tiered object + high-performance file storage ensures low-latency model weight loading and cost-effective bulk data retention.

- LOW** Cost: Object storage is commodity-priced; tiering to cold storage further reduces cost
- LOW** Complexity: Fully managed; standard SDK access across all clouds
- MEDIUM** Vendor lock-in: Egress costs make migration expensive; format standards (Parquet, Delta) reduce risk
- LOW** Time to value: Ready to use within hours; no provisioning lead time



Data Platform

 82/100

Confidence: 88%

RECOMMENDED

Azure Databricks + Azure Data Factory

Unified data lakehouse pattern enables batch ETL, real-time streaming, and governed data products that feed AI pipelines.

- MEDIUM** Cost: Lakehouse compute scales with query volume; optimize with scheduled jobs
- HIGH** Complexity: Requires data engineering expertise for pipelines, governance, and quality
- MEDIUM** Vendor lock-in: Open formats (Delta Lake, Apache Iceberg) preserve portability
- HIGH** Time to value: Data readiness is typically the critical path; allow 8–12 weeks for initial pipelines

Model Layer

 85/100

Confidence: 88%

RECOMMENDED

Azure OpenAI Service (GPT-4o, o1) + Azure ML

Managed API endpoints minimize ops overhead and enable rapid iteration.

- MEDIUM** Cost: Token-based pricing scales linearly; monitor usage to avoid runaway costs
- LOW** Complexity: REST API integration; no model serving infrastructure to manage
- HIGH** Vendor lock-in: API surface, pricing, and model versions are controlled by the provider
- LOW** Time to value: API key and quota approval is the only setup step; hours to first call

RAG / Vector Layer

 85/100

Confidence: 88%

RECOMMENDED

Azure AI Search (integrated vectorization) + Cosmos DB

Retrieval-augmented generation grounds model responses in your proprietary knowledge base, reducing hallucination and improving accuracy.

- LOW** Cost: Vector storage is low cost; embedding generation is the main operational expense
- MEDIUM** Complexity: Chunking strategy, embedding model selection, and reranking require iteration
- MEDIUM** Vendor lock-in: Open embedding standards (OpenAI-compatible) enable provider switching
- MEDIUM** Time to value: Initial RAG pipeline in 2–3 weeks; tuning for accuracy takes longer

Orchestration / Agent

 85/100

Confidence: 88%

RECOMMENDED

Azure AI Foundry Agent SDK + LangChain

Multi-step agent workflows with tool calling and human-in-the-loop checkpoints enable complex enterprise automation patterns.

- LOW** Cost: Orchestration frameworks are open source; cost is primarily developer time
- HIGH** Complexity: Agent state management, tool calling, and HITL require careful design
- MEDIUM** Vendor lock-in: LangChain/LangGraph are open; cloud-native agent frameworks add lock-in
- HIGH** Time to value: Complex agent workflows take 4–8 weeks to production-harden



Observability

 82/100

Confidence: 92%

RECOMMENDED

Azure Monitor + Application Insights

Full-stack observability with LLM-specific metrics (token cost, latency, hallucination rate) and distributed tracing for AI pipelines.

- LOW** Cost: Log/metric ingestion cost scales with volume; sample rates reduce cost
- LOW** Complexity: Standard instrumentation libraries; LLM-specific SDKs add minimal overhead
- MEDIUM** Vendor lock-in: OpenTelemetry standard reduces switching cost across observability providers
- LOW** Time to value: Baseline dashboards deployable in 1–2 weeks; LLM metrics need custom work

Security

 82/100

Confidence: 92%

RECOMMENDED

Microsoft Entra ID + Defender for AI + Purview

Enterprise security baseline with RBAC, MFA, and audit trails across all AI workloads.

- MEDIUM** Cost: Security baseline is included in cloud landing zone; additional controls are incremental
- HIGH** Complexity: Security architecture spans identity, network, data, and AI-specific controls
- MEDIUM** Vendor lock-in: Security tooling is portable; SIEM and CSPM support multi-cloud
- MEDIUM** Time to value: Security baseline deployable in 2–3 weeks; compliance certification takes months

Identity

 81/100

Confidence: 92%

RECOMMENDED

Microsoft Entra ID (OIDC / SSO)

Federated identity with SAML/OIDC ensures single sign-on across all AI factory components with centralized RBAC policy enforcement.

- LOW** Cost: Federated identity is included in enterprise cloud agreements
- LOW** Complexity: SAML/OIDC standards are well-understood; SSO integration is routine
- MEDIUM** Vendor lock-in: OIDC standard ensures portability; proprietary MFA methods add lock-in
- LOW** Time to value: SSO federation typically completed in 1–2 weeks

API Layer

 84/100

Confidence: 88%

RECOMMENDED

Azure API Management

Centralized API management provides rate limiting, authentication, usage analytics, and developer portal for internal and external consumers.

- LOW** Cost: API management is a minor cost relative to compute; included in enterprise tiers
- LOW** Complexity: Standard configuration; rate limiting and auth policies are template-driven
- MEDIUM** Vendor lock-in: OpenAPI specification standard preserves portability of service contracts
- LOW** Time to value: API gateway configuration is same-day; developer portal takes 1–2 weeks



Network / Connectivity



Confidence: 88%

RECOMMENDED

Azure Virtual Network + ExpressRoute + Private Endpoints

Private connectivity eliminates public internet exposure for AI inference traffic. Private endpoints ensure all traffic stays within the cloud backbone.

- HIGH** Cost: Dedicated circuits (ExpressRoute / Direct Connect) carry monthly port fees of \$2–10K+
- MEDIUM** Complexity: BGP routing, private endpoints, and firewall rules require network engineering
- LOW** Vendor lock-in: Physical circuit contracts are with telco carriers, not cloud providers
- MEDIUM** Time to value: Dedicated circuit provisioning takes 4–12 weeks; cloud-native VPN is faster